

Unit-1: Fundamentals of Data Analytics

1.2 Understanding the Data

1.2.1 Quantitative Data: Discrete and Continuous

1.2.2 Qualitative Data: Non-numerical (Normal and Ordinal)

1.2.1 Quantitative Data: Discrete and Continuous

Quantitative data refers to **numerical** data that can be **measured or counted**.

It answers the questions “**How much?**” or “**How many?**” and consists of values that can be analyzed using mathematical tools and algorithms.

This type of data is often **structured**, making it suitable for tasks such as **trend forecasting, statistical analysis, budgeting, pattern identification, and performance measurement**.

➤ Types of Quantitative Data:

1. Discrete Data:

- **Definition:** Data that can take **only specific, distinct values** (usually whole numbers).
- **Nature:** *Countable* — there are no intermediate or fractional values between two data points.
- **Examples:**
 - Number of students in a class (e.g., 25, 30)
 - Number of cars in a parking lot
 - Number of products sold
 - Test scores (if given as whole numbers)

2. Continuous Data:

- **Definition:** Data that can take **any value within a given range**.
- **Nature:** *Measurable* — can include fractions or decimals.
- **Examples:**
 - Height of a person (e.g., 5.6 feet)
 - Temperature (e.g., 37.2°C)
 - Time (e.g., 2.45 hours)

➤ Key Differences:

Feature	Discrete Data	Continuous Data
Type of values	Specific, fixed values	Any value in a range

Feature	Discrete Data	Continuous Data
Nature	Countable	Measurable
Examples	Number of books, Test Scores	Weight, distance, time, temperature
Can be fractional?	No	Yes

1.2.2 Qualitative Data: Non-Numerical (Nominal and Ordinal)

Qualitative data refers to **non-numerical information** that describes categories, labels, characteristics, or experiences rather than quantities.

It answers the questions “**What kind?**” or “**Which category?**”, capturing aspects that **numbers cannot measure directly**.

Examples include **customer feedback, product reviews, interview transcripts, and social media comments**.

➤ Forms of Qualitative Data

- **Structured:** Organized and coded (e.g., survey responses with predefined options).
- **Unstructured:** Free-text, descriptive, or narrative (e.g., interview transcripts, open-ended feedback).

➤ Types of Qualitative Data:

1. Nominal Data:

- **Definition:** Data that represents Categories without any inherent order, hierarchy or ranking.
- **Nature:** Labels only — can be counted, but not logically ordered.
- **Examples:**
 - Gender (male, female)
 - Blood group (A, B, AB, O)
 - Eye color (blue, green, brown)
 - Nationality (Indian, American, etc.)
- **Key Point:** We can **name and count** them, but **not rank** them.

2. Ordinal Data:

- **Definition:** Data that represents categories with a meaningful order or ranking, but the **differences between values are not measurable or uniform**.
- Ranking exists, but intervals are not equal or known.
- **Nature:** Order matters, but spacing between levels is inconsistent.
- **Examples:**
 - Education level (High School < Bachelor's < Master's < Ph.D.)
 - Customer satisfaction (Very dissatisfied < Neutral < Very satisfied)

- Class rank (1st, 2nd, 3rd)
- Survey rating (1 = Poor, 5 = Excellent)
- **Key Point: Ranking exists, but intervals are not equal.**

➤ **Key Differences:**

Feature	Nominal Data	Ordinal Data
Order/Ranking	No	Yes
Type of values	Categories only	Ordered categories
Examples	Gender, Blood group, Colors	Satisfaction level, Education Ranks, Survey ratings
Measurement	Labeling & Counting	Ranking without measurable intervals

Common Use Cases of Qualitative Data

- Understanding **customer behavior**
- Analyzing **market trends**
- Studying **user experiences**
- Organizing **feedback and opinions** for improvement

Summary Chart

Data Type	Sub-type	Characteristics	Examples
Quantitative	Discrete	Countable, whole numbers	Number of pets or students
	Continuous	Measurable, includes decimals	Height, weight, temperature
Qualitative	Nominal	Categorical, no order	Gender, nationality, colors
	Ordinal	Categorical with order, no fixed scale	Satisfaction levels, class ranks

1.1 Exploratory Data Analysis (EDA)

1.1.1 Types of Exploratory Data Analysis:

1.1.2 Univariate Analysis

1.1.3 Bivariate Analysis

1.1.4 Multivariate Analysis

1.1.5 Handling Missing Data and Outliers

Definition:

Exploratory Data Analysis (EDA) is the **first step in data analysis**. It is the process of analyzing datasets to **summarize their main characteristics**, often using visual methods. EDA helps

understand the **structure, patterns, and anomalies** in data before applying statistical models or machine learning techniques.

Goal:

To **understand the data** and prepare it for further analysis.

Objectives of EDA

It helps in:

- Detecting **patterns and trends** in the data.
- Identifying **missing values**.
- Spotting **outliers and anomalies**.
- Choosing the **right analysis techniques**.
- Summarizing data through **statistics and visualizations**.

1.1.1 Types of Exploratory Data Analysis

EDA can be categorized in **two different ways**:

1. **By the Scope of Variables**

- **Univariate Analysis** → examines a single variable (e.g., distribution, frequency, summary stats).
- **Bivariate Analysis** → examines the relationship between two variables (e.g., correlation, scatter plots).
- **Multivariate Analysis** → Examines interactions among more than two variables (e.g., regression, PCA, clustering).

2. **By the Techniques Used**

- **Descriptive Statistics** → Summarizing data using measures like mean, median, standard deviation, variance.
- **Visualization Techniques** → Using charts and plots (histograms, boxplots, scatter plots, heatmaps) to identify patterns.
- **Comparative Analysis** → Comparing groups or datasets (e.g., t-tests, ANOVA).
- **Inferential Analysis** → Drawing conclusions about a population based on sample data (e.g., hypothesis testing, confidence intervals).

1.1.2 Univariate Analysis

Univariate analysis focuses on **studying a single variable** to understand its characteristics, distribution, and patterns. It is the simplest form of data analysis and helps describe the central tendency and spread of the data.

Purpose

- Understand the **distribution** of a variable.
- Describe **central tendency** (mean, median, mode).

- Identify **spread** (variance, standard deviation).
- Detect **outliers** (using boxplots).

Techniques

1. Numerical Variables

- **Measures**

Measure	Definition	Example
Mean	Average of all values.	Ages of students: 10, 12, 14 → Mean = $(10+12+14)/3 = 12$
Median	Middle value when data is sorted.	Ages: 10, 12, 14 → Median = 12
Mode	Most frequently occurring value.	Ages: 10, 12, 12, 14 → Mode = 12
Variance	Average of squared differences from the mean.	Ages: 10, 12, 14 → Variance = $((10-12)^2 + (12-12)^2 + (14-12)^2)/3 = 2.67$
Standard Deviation	Square root of variance; measures data spread.	$\sqrt{2.67} \approx 1.63$

- **Visualizations**
 - **Histogram** → Shows data distribution.
 - **Box Plot** → Detects outliers and shows spread.
 - **Density Plot** → Smooth estimate of data distribution.

2. Categorical Variables

- **Techniques**
 - **Frequency tables** → Count occurrences of each category.
 - **Bar Chart / Pie Chart** → Visual representation of category counts.
- **Examples**
 - **Numerical:** Age distribution of students in a class (Histogram, Box Plot).
 - **Categorical:** Count of different genders in a class (Bar Chart, Pie Chart)
- **Key Points**
 - Focuses on **one variable at a time**.
 - Helps to **understand distribution, central tendency, and spread**.
- **Tools**
 - **Continuous data:** Histogram, Boxplot, Summary statistics.
 - **Categorical data:** Frequency table, Bar chart, Pie chart.

1.1.3 Bivariate Analysis

Bivariate Analysis focuses on **analyzing the relationship between two variables** to find connections, correlations, and dependencies. It helps understand **how one variable changes with respect to another** and is essential for exploring associations in data.

Purpose

- Identify relationships or associations between two variables.
- Understand interactions, correlations, or dependencies.
- Detect patterns, trends, and anomalies.

Types of Variable Combinations and Techniques

Variable Type	Techniques	Description	Example
Numerical vs Numerical	Scatter Plot, Correlation Coefficient, Covariance	Scatter plots visualize relationship; correlation measures strength and direction; covariance shows how two variables vary together.	Study hours vs Exam scores
Categorical vs Categorical	Cross-tabulation, Stacked/Grouped Bar Chart, Chi-square test	Frequency distribution of two categorical variables to examine association.	Distribution of gender across departments
Numerical vs Categorical	Box Plot, Violin Plot, Grouped Bar Chart	Compare distribution of numerical variable across categories.	Exam scores vs Grade levels

Key Techniques

1. **Scatter Plot**
 - Visualizes relationship between two continuous variables.
 - Helps detect patterns, trends, or clusters.
2. **Correlation Coefficient (Pearson's r)**
 - Measures strength and direction of linear relationship between two numerical variables.
 - Range: -1 to +1.
 - Positive $\rightarrow$ variables increase together
 - Negative $\rightarrow$ one increases, other decreases
 - 0 $\rightarrow$ no linear relationship
3. **Covariance**

- Measures how two variables change together.
- Not standardized → use correlation for clearer interpretation.
- 4. **Cross-tabulation / Contingency Table**
 - Frequency table for two categorical variables.
 - Can be used with Chi-square test for statistical significance.
- 5. **Box Plot / Violin Plot**
 - Compare numerical variable distributions across categories.
 - Show median, quartiles, outliers, and density.
- 6. **Line Graphs**
 - Useful for **time series data** to track trends of two variables over time.

Examples

- Numerical vs Numerical: Relation between study hours and exam scores (Scatter Plot, Correlation).
- Categorical vs Categorical: Distribution of gender across departments (Grouped Bar Chart, Cross-tab).
- Numerical vs Categorical: Exam scores across different grade levels (Box Plot).

Key Points

- Analysis of **two variables together**.
- Purpose: Identify relationships or associations.
- Types and Tools:
 - Continuous vs Continuous → Scatter plot, Correlation, Covariance
 - Categorical vs Categorical → Cross-tabulation, Stacked/Grouped Bar Chart, Chi-square test
 - Categorical vs Continuous → Box Plot, Violin Plot, Grouped Bar Chart

1.1.4 Multivariate Analysis

Multivariate Analysis focuses on **analyzing relationships among three or more variables simultaneously**. It helps understand **complex interactions and dependencies** in the data and is essential for **statistical modeling, predictive analytics, and dimensionality reduction**.

Purpose

- Understand relationships among multiple variables.
- Identify interactions and dependencies.
- Reduce complexity in large datasets.
- Useful for **modeling, pattern discovery, and prediction**.

Techniques

Technique	Description	Example
Heatmap	Visualizes correlations between multiple numerical variables.	Correlation between age, income, and education level.
Pair Plot (Scatterplot Matrix)	Plots scatter plots for all variable pairs, showing interactions.	Exploring relationships between study hours, exam score, and attendance.
Principal Component Analysis (PCA)	Reduces dimensionality of large datasets while retaining important information.	Reducing a dataset with many features to 2–3 principal components for visualization.
Multivariate Regression / Classification Models	Analyzes the effect of multiple predictors on a target variable.	Effect of study hours, gender, and family income on exam performance.
Cluster Analysis	Groups similar observations based on multiple variables.	Segmenting customers based on age, income, and spending habits.
Spatial Analysis	Visualizes and analyzes geographical data using maps.	Distribution of sales across different cities.
Time Series Analysis	Analyzes datasets with time-based data to identify trends and patterns. Techniques include line plots, autocorrelation, moving averages, and ARIMA models.	Predicting monthly sales using historical data.

Examples

- Exploring how **age, education level, and income** relate to each other.
- Using **PCA** to reduce dimensions in a dataset with many features.
- Evaluating the combined effect of **study hours, gender, and family income on exam performance**.
- Mapping **geographical sales distribution** using spatial analysis.

- Predicting future sales with **time series analysis**.

Key Points

- Focuses on **three or more variables**.
- Purpose: Understand **complex interactions** and **dependencies**.
- Techniques include:
 - Heatmap
 - Pair Plot
 - PCA
 - Multivariate Regression / Classification
 - Cluster Analysis
 - Spatial Analysis
 - Time Series Analysis

1.1.5 Handling Missing Data and Outliers

Real-world datasets often have **missing values** or **outliers** that can skew results. EDA involves identifying and managing these issues.

Real-World Data Challenges

In real-world scenarios, data is often **dirty, incomplete, noisy, or inconsistent**, making cleaning and preprocessing essential before analysis.

Common Data Quality Issues

Type	Description	Example
Incorrect Data	Errors from faulty instruments, human/computer mistakes, or transmission faults.	Salary = “-10”
Incomplete Data	Missing attribute values or records.	Occupation = “ ”
Noisy Data	Random errors or outliers present.	Measurement device malfunction
Inconsistent Data	Conflicting or incompatible entries.	Age = “42”, Birthday = “03/07/2010”
Intentional Errors	Disguised missing data or defaults.	Jan. 1 as everyone’s birthday

➤ Handling Missing Data:

Detection

- Use tools like `isnull()` in Python or summary tables to identify missing values.

Causes

- Equipment malfunction
- Data not entered or deleted due to inconsistency
- Misunderstanding during data entry
- Data once considered unimportant
- Historical data not recorded

Problems

- Bias in analysis
- Reduced accuracy of models
- Some algorithms cannot handle missing data directly

Techniques to Handle Missing Data

Method	Description
Ignore the Tuple	Drop records with missing values (not ideal if large % missing).
Manual Fill	Fill missing data manually (time-consuming, impractical).
Global Constant	Replace with “Unknown” or “Missing”.
Mean/Mode Imputation	Use attribute mean or mode for replacement.
Class-Based Mean/Mode	Replace using mean/mode of records from the same class.
Predictive Imputation	Estimate missing values using regression, KNN, or Bayesian models.
Random Imputation	Randomly sample from the data’s distribution — maintains spread but may be unrealistic.
Advanced Estimation	Use inference-based or model-based replacement (e.g., Decision Trees).
Leave as is	(some algorithms handle missing values)

Example

If “cubic inches” (engine size) is missing:

- American car → ~300 cubic inches
- Japanese car → ~100 cubic inches Predict replacement based on other attributes like origin or horsepower.

➤ Handling Outliers:

Detection

- **Graphical:** Box plot, histogram, scatter plot
- **Statistical:**
 - **Z-score:** >3 or <-3 indicates an outlier
 - **IQR Method:** Data outside $1.5 \times \text{IQR}$ range
- Outliers may represent:
 - **Errors** (e.g., impossible values)
 - **True but rare cases** (shouldn't always be removed)

Techniques

Method	Description
Remove Outliers	Delete extreme or erroneous values.
Capping (Winsorizing)	Limit extreme values to a threshold.
Transform Data	Apply log or square-root transformations to reduce skewness.
Cluster/Regression Methods	Detect and treat extreme cases systematically.

Example

Histogram of vehicle weights shows one extreme outlier (192.5 pounds).
→ Check validity — likely data entry error.

Key Insights

- **Deleting records is not always ideal** — may bias the dataset.
- **Imputation should maintain statistical integrity** (mean, variance, and distribution).
- **Outliers and missing values should be examined contextually** — not all “odd” data is wrong.
- **Clean, consistent data ensures better model performance** and meaningful insights.

➤ Handling Noisy Data

Noise Sources

- Data entry/transmission errors
- Faulty sensors or measurement tools
- Inconsistent naming conventions

Noise Reduction Methods

Technique	Description
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Technique	Description
Binning	Sort and group into equal-frequency bins; smooth by mean, median, or boundaries.
Regression	Fit data using regression functions to smooth trends.
Clustering	Detect and remove or correct outliers that don't belong to any cluster.
Combined Inspection	Computer detects suspicious values, human confirms/corrects.

Summary Table

Step	Purpose	Tools/Techniques
Univariate	Understand one variable	Histograms, Bar charts, Summary stats
Bivariate	Explore relationship between two	Scatter plots, Correlation, Box plots
Multivariate	Understand complex relationships	Heatmaps, PCA, Multivariate regression
Missing Data	Handle gaps in data	Imputation, Deletion, Prediction
Outliers	Detect and treat unusual values	Box plot, Z-score, IQR

1.3 Spread of Data

1.3.1 Normal Distribution

1.3.2 Skewed Distribution

1.3.3 Skewness and Kurtosis

The spread of data (also known as **variability** or **dispersion**) refers to how much the data values differ or vary from the central value — such as the **mean**, **median**, or **mode**. Understanding the spread helps interpret the **shape**, **distribution**, and **behavior** of a dataset, making it easier to **detect patterns**, **outliers**, or **irregularities**.

1.3.1 Normal Distribution

➤ Definition:

A **normal distribution**, also known as the **Gaussian distribution** or **bell curve**, is a **symmetrical, bell-shaped curve** where most data values **cluster around the mean**, and the **probabilities taper off equally on both sides**.

➤ **Key Characteristics:**

- **Mean = Median = Mode**
- Symmetrical shape: **Symmetric** about the mean (both sides mirror each other)
- **Fully described by two parameters:**
 - Mean (μ): Center of the curve
 - Standard Deviation (σ): Spread or dispersion of the curve
- **Empirical Rule (68–95–99.7 Rule):**
 - ~68% of data lies within $\pm 1\sigma$
 - ~95% within $\pm 2\sigma$
 - ~99.7% within $\pm 3\sigma$
- **Area under the curve = 1** (represents total probability)
- Commonly used in natural and social phenomena — **heights, IQ scores, test results**

➤ **Visualization:**

A smooth, **bell-shaped curve** centered on the mean (μ), with tails decreasing symmetrically on both sides.

1.3.2 Skewed Distribution

➤ **Definition:**

A skewed distribution is one in which data is not symmetrical — **it has a longer tail on one side of the mean**.

➤ **Types of Skewness**

Type	Description	Shape/Example
Right-Skewed (Positive Skew)	Tail extends to the right ; most data lies on the left. Mean > Median > Mode.	Income, house prices, waiting times
Left-Skewed (Negative Skew)	Tail extends to the left ; most data lies on the right. Mean < Median < Mode.	Age at retirement, easy exam scores

➤ **Impact on Central Tendency**

- **Right Skew:** Mean > Median
- **Left Skew:** Mean < Median

➤ **Visualization:**

- Right Skew: Curve peaks on the left, tail stretches right
- Left Skew: Curve peaks on the right, tail stretches left

➤ **Importance**

- Many statistical tests (e.g., regression, t-tests) assume **normality**.
- Highly skewed data can **distort averages and variance**.
- **Transformation** (log, square root, etc.) can reduce skewness for accurate modeling.

1.3.3 Skewness and Kurtosis

A. Skewness

- **Definition:** Skewness measures the **asymmetry** of the distribution around its mean. It tells whether data values are more concentrated on one side of the mean.

Skewness Value	Interpretation	Shape
0	Perfectly symmetrical (Normal)	Balanced bell curve
> 0	Right-skewed (Positive Skew)	Tail to the right
< 0	Left-skewed (Negative Skew)	Tail to the left

- **Formula (conceptually)**

$$Skewness = \frac{\sum (x_i - \bar{x})^3}{n \times s^3}$$

Where $\bar{x}$ = mean, s = standard deviation.

B. Kurtosis

- **Definition:** Kurtosis measures the “**tailedness**” or **peakedness** of a distribution — how heavy or light its tails are compared to the normal distribution.

➤ **Types of Kurtosis:**

Types of Kurtosis

Type	Description	Kurtosis Value (Excess Kurtosis)
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Type	Description	Kurtosis Value (Excess Kurtosis)
Mesokurtic	Normal tails (reference distribution)	≈ 0
Leptokurtic	Heavy tails, sharp peak (more outliers)	> 0
Platykurtic	Light tails, flat peak (fewer outliers)	< 0

➤ Interpretation

- **Leptokurtic** → Sharp central peak, heavy tails (higher chance of outliers).
- **Platykurtic** → Flatter peak, lighter tails (data spread out evenly).
- **Mesokurtic** → Similar to normal distribution (baseline).

➤ Why It Matters

- **High kurtosis** → More extreme values (outliers), heavier tails, sharper peak.
- **Low kurtosis** → Fewer extreme values, lighter tails, flatter peak.
- Indicates **risk of outliers** and how concentrated data is around the mean.

➤ Formula (Excess Kurtosis)

$$Kurtosis_{excess} = Kurtosis - 3$$

(So that a normal distribution has an excess kurtosis of **0**.)

Summary Table

Concept	Description	Shape / Example
Normal Distribution	Symmetrical bell-shaped curve; mean = median = mode	Heights, IQ scores
Right Skewed Distribution	Tail to the right; mean > median	Income, waiting times
Left Skewed Distribution	Tail to the left; mean < median	Retirement age, easy exams
Skewness	Measures asymmetry of distribution	0 = symmetric; >0 = right skew; <0 = left skew
Kurtosis	Measures tail weight and peak sharpness	>0 = sharp peak; <0 = flat peak

Key Takeaways

- **Normal distribution** serves as the foundation for statistical analysis.
- **Skewness** and **kurtosis** quantify **deviations from normality**.
- Real-world data often deviates due to external factors — recognizing these deviations helps improve data modeling accuracy.